

Arash Bahari Kordabad

Curriculum Vitae

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Postdoctoral Research Fellow, Max Planck Institute for Software Systems (MPI-SWS)

Kaiserslautern, Germany [LinkedIn](#)

Academic Profile

My research combines reinforcement learning, model predictive control, mathematical optimization, stochastic systems, and formal methods to develop reliable, interpretable, and data-efficient decision-making algorithms with explicit guarantees under uncertainty. Applications include autonomous vehicles, robotics, aerospace and maritime systems, smart grids, energy networks, and industrial processes.

Academic Career

Postdoctoral Research Fellow, MPI-SWS, Kaiserslautern, Germany May 2023–present

Research on safe autonomy, stochastic reachability, distributionally robust optimization, temporal logic, and multi-agent awareness and control.

Visiting Researcher, Aalborg University, Denmark

Nov 2021–Aug 2022

Department of Electronic Systems. Safe reinforcement learning and distributionally robust optimization.

Host: Prof. Rafal Wisniewski.

Education

Ph.D. in Engineering Cybernetics, NTNU

Feb 2020–Apr 2023

Thesis: *Theoretical Properties of Learning-Based MPC*, repository link. Supervisor: Prof. Sébastien Gros; co-supervisor: Prof. Anastasios Lekkas.

M.Sc. in Mechanical Engineering, Sharif University of Technology

Sep 2017–Sep 2019

Thesis: *Control of Bifurcation and Chatter Suppression in Peripheral Milling Process*. Supervisor: Prof. Hamed Moradi. GPA: 19.41/20.

B.Sc. in Mechanical Engineering, University of Tabriz

Sep 2013–Sep 2017

Thesis: *On the Muscle Models as Viscoelastic Material and Comparison of Force-Length Models for Active Skeletal Muscle*. Supervisor: Prof. Kamal Jahani. GPA: 18.1/20; ranked fifth among 99 students.

Research Interests

Markov decision processes; reinforcement learning; economic, robust, stochastic, and distributionally robust MPC; second-order and quasi-Newton learning; Q-learning and dissipativity; data-driven control; stochastic reachability and Lyapunov methods; polynomial and sum-of-squares optimization; signal temporal logic; control barrier functions; multi-agent planning; risk-aware decision making.

Honors and Awards

- **2026:** Engineering Applications of Artificial Intelligence Paper Prize Award, with Wenqi Cai and Sébastien Gros, for *Energy Management in Residential Microgrid Using Model Predictive Control-Based Reinforcement Learning and Shapley Value* (2023).
- **2018:** First rank in Dynamics and Control (GPA 19.41/20), and third among all Mechanical Engineering students at Sharif University of Technology.
- **2017:** Selected among the top 40 Mechanical Engineering students nationwide for the Iranian Scientific Olympiad for University Students.
- **2017:** Ranked fifth among 99 B.Sc. Mechanical Engineering students, University of Tabriz.
- **2013:** Ranked 669th among more than 252,000 participants in the Iranian national university entrance examination (approximately the top 0.27%).
- **2013:** Honorary diploma, International Mathematics Tournament of Towns.

Research Output and Selected Publications

13 journal papers (published or accepted), 4 submitted manuscripts, and 20 conference papers; 25 first-author records. The complete publication list is maintained on my website.

1. A. Valaei, **A. Bahari Kordabad**, S. Soudjani, “Second-Order Policy Gradient Methods for the Linear Quadratic Regulator.” *Engineering Applications of Artificial Intelligence*, accepted, 2026. [arXiv](#)
2. **A. Bahari Kordabad**, R. Majumdar, H. J. Motwani, S. Soudjani, “On Certificates for Almost Sure Reachability in Stochastic Systems.” *IEEE Transactions on Automatic Control*, 2026. DOI [arXiv](#)
3. **A. Bahari Kordabad**, M. Zanon, S. Gros, “Equivalence of optimality criteria for Markov decision process and model predictive control.” *IEEE Transactions on Automatic Control*, 2024. DOI [arXiv](#)
4. W. Cai, **A. Bahari Kordabad**, S. Gros, “Energy Management in Residential Microgrid Using Model Predictive Control-Based Reinforcement Learning and Shapley Value.” *Engineering Applications of Artificial Intelligence*, vol. 119, art. 105793, 2023; EAAI Paper Prize Award 2026. DOI [arXiv](#)
5. **A. Bahari Kordabad**, R. Wisniewski, S. Gros, “Safe reinforcement learning using Wasserstein distributionally robust model predictive control.” *IEEE Access*, 2022. DOI [arXiv](#)
6. **A. Bahari Kordabad**, D. Brandner, S. Gros, S. Lucia, S. Soudjani, “Quasi-Newton Compatible Actor-Critic for Deterministic Policies.” *European Control Conference (ECC)*, 2026. Publisher [arXiv](#)

Research Projects and Industrial Collaboration

Auto-CyPheR, ERC-funded

Sep 2025–Sep 2027

Postdoctoral research on autonomous cyber-physical systems, learning-enabled decision making under uncertainty, and formal guarantees.

SymAware, EIC-funded

May 2023–Sep 2025

Research on awareness and decision making in multi-agent systems. Partners include MPI-SWS, KTH, Uppsala University, Eindhoven University of Technology, Siemens Digital Industries Software, and the Netherlands Aerospace Centre (NLR).

SARLEM, NTNU

2020–2023

Safe Reinforcement Learning using Model Predictive Control. Research Council of Norway grant NFR 300172; project identifier UV988962100. Industrial partners include DNV and Kongsberg Maritime.

Industry-supported M.Sc. research, Tehran

2017–2019

Modeling, control, and chatter suppression in milling processes; partially supported by a machinery company. Team: one professor and three students.

Teaching Experience

Doctoral educational contributions, NTNU, Norway **2021–2022**

Advanced Nonlinear Systems (TK8103, spring 2021): teaching a section on stability of perturbed systems using slides, illustrations, and handwritten proofs. Topics in System and Control Theory (TK8111, autumn 2022): preparing a concise reinforcement-learning textbook for a self-study course, evaluated by a committee.

Teaching Assistant, Sharif University of Technology, Iran **2017–2019**

Automatic Control; Mechanical Vibrations (3 credits, 2018–2019).

Teaching Assistant, University of Tabriz, Iran **2014–2016**

Calculus I and II (3+3 credits), Ordinary Differential Equations (3 credits), and Engineering Mathematics (3 credits).

Mathematics teacher and course coordinator, Kanoon Farhangi Amoozesh, Tehran **2017–2019**

High-school mathematics teaching, educational consulting, and course coordination. Employment dates: November 2017–November 2019.

Mathematics teacher, SAYERI Private Educational Institution, Tabriz **2015–2017**

High-school mathematics teaching over two years (four semesters).

Supervision and Mentoring

At MPI-SWS, I have supervised research interns including Andrea Da Col and Amirreza Valaei, working on learning, optimization, and autonomous systems. Andrea subsequently became a Ph.D. researcher at KTH. At NTNU, I mentored junior Ph.D. researchers through research discussions, literature review, and feedback on their scientific work.

Pedagogical and Professional Training

- Basic University Didactics (IDT8001), NTNU, spring 2021.
- Research Ethics and Research-Based Innovation, NTNU.
- Seminar on the presentation of educational and scientific material, Sharif University of Technology.

Selected Invited and Educational Talks

- **July 2026, ECC, Reykjavik, Iceland:** “Quasi-Newton Compatible Actor-Critic for Deterministic Policies.”
- **December 2025, SESAR Innovation Days, Bled, Slovenia:** “Intent-Aware MPC for Aircraft Detect-and-Avoid with Response Delay: A Comparative Study with ACAS Xu.”
- **June 2025, ECC, Thessaloniki, Greece:** “Robust Model Predictive Control for Aircraft Intent-Aware Collision Avoidance.”
- **2024, CDC, Milan, Italy:** “Distributionally Robust Control for Chance-Constrained Signal Temporal Logic Specifications.”
- **July 2023, IFAC World Congress, Japan:** “Reinforcement Learning for MPC: Fundamentals and Current Challenges.” Open invited session: *Recent Advances in Automated Learning and Calibration of MPC Policies*; audience of more than 75 participants.
- **February 2023, TU Eindhoven, Netherlands:** “Intersection of Reinforcement Learning and MPC.” Host: Prof. Dinesh Krishnamoorthy.

Academic Service

Reviewer for CDC, ACC, ECC, and NMPC, and for IEEE Transactions on Automatic Control, IEEE Open Journal of Control Systems, ISA Transactions, and Engineering Applications of Artificial Intelligence. Guest editor of *Advanced Control Strategies for Autonomous Maritime Systems*, Journal of Marine Science and Engineering.